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On-line monitoring of damage to a sintering press

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Abstract

A manufacturer of high strength cutting tools utilizes an SPM-75 sintering press to manufacture custom designed blade segments. The sintering press uses graphite electrodes, brass plates and brass heads to pressurize, heat and join the blade segments. It was found that graphite electrodes periodically oxidize and begin to warp, which in turn warps the brass plates adjacent to the electrodes. The warping of the plates damages the brass head units adjoining the plates. It is recommended that the brass plates be resurfaced after two days if production has been intensive and no more than five days if production has been light to reduce wear on the electrodes and the warping of the brass head units. © 1998 Elsevier Science Ltd. All rights reserved.

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1. Introduction

Cushion Cut, a division of Boart Longyear Company, is a manufacturer of high strength cutting tools. Their main line of products consists of blades and bits for cutting concrete and asphalt. Their line of tools includes bits with diamond composite tips that are used in deep drilling worldwide.

Cushion Cut uses a Dr Fritsch manufactured hot press in their production line. It consists of a dewaxing station, station 1, whose temperature does not exceed 500°C and a pressing station, station 2. In station 2, temperatures are maintained between 800–1200°C, with pressures up to 2000 psi. The hydraulic press components consist of two brass head units, two brass plates and a pair of graphite electrodes, Fig. 1. These dissimilar materials are pressed against each other, and under extreme pressures and high temperatures they crack. When the electrodes crack, they are replaced relatively quickly. However, when the brass heads crack, this causes unscheduled down-

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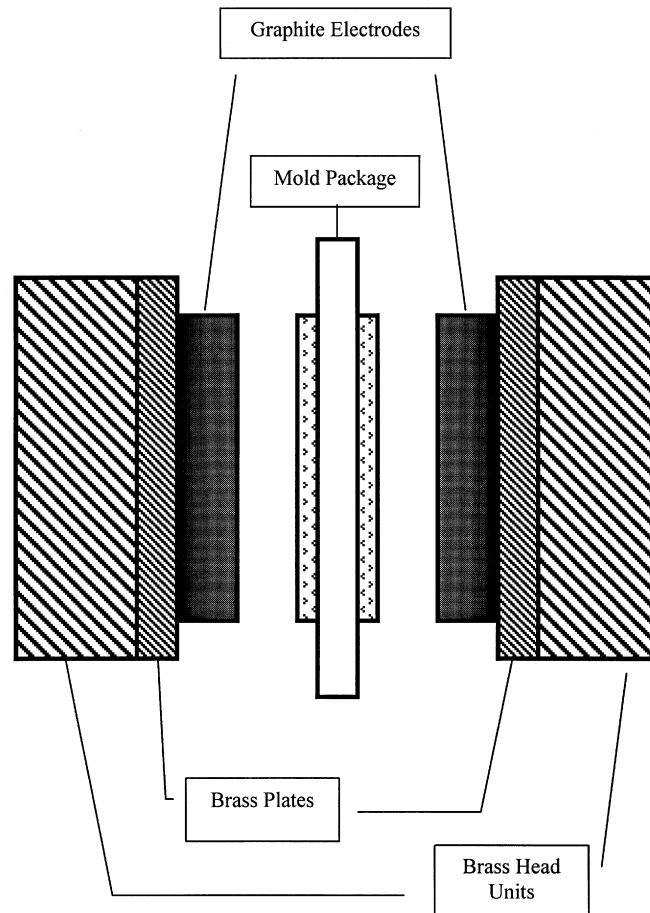
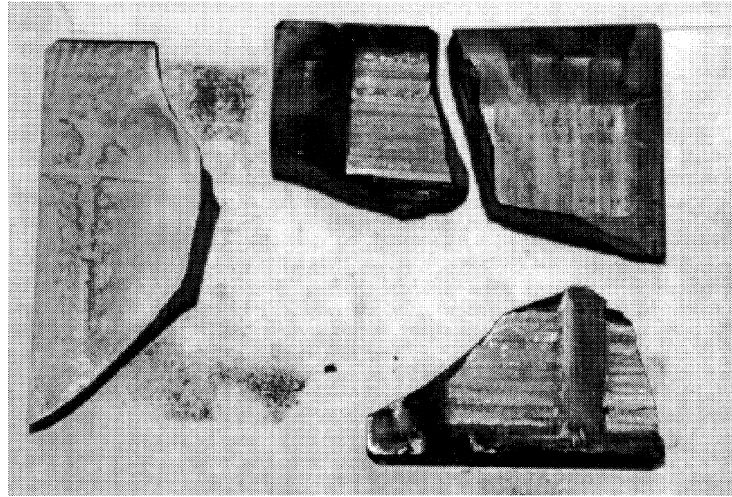


Fig. 1. Sintering press pressure components and the mold package.

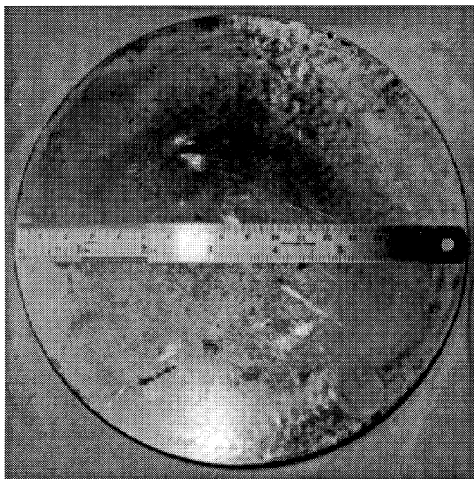
times of 3–4 h per replacement, which can result in thousands of dollars of losses. The objective of this study is to determine the cause of these failures or how to prevent them.

2. Background

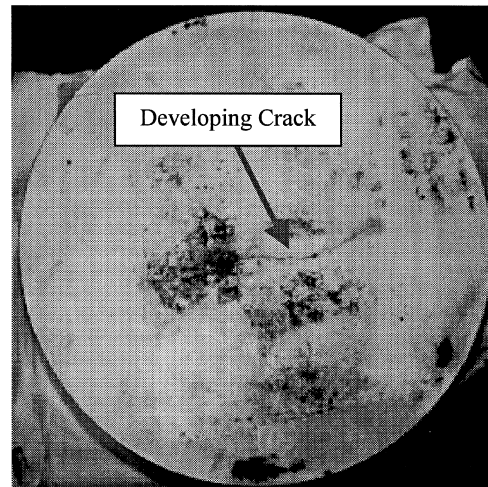
A Dr Fritsch SPM 75 hot press is being used to press diamond blade segments at temperatures between 800°C and 1200°C and pressures near 2000 psi [1]. Too often, the three main components of the press are either cracking or deteriorating into small fragments. One of the main components is the graphite electrode, which after repeated use at the high temperatures and pressures, wears at the edges and breaks down into pieces between one-quarter and one-tenth of the original size. The second main component is the brass plate, which serves as a protective medium between the brass head and the graphite electrode. The brass head, the third main component, has water lines



Pieces of a Graphite Electrode



Brass Plate with surface damage.



Brass Head with surface damage and fracture.

Fig. 2. The three main components and their failures are shown. Graphite electrodes often break into pieces $\frac{1}{4}$ their original size. The brass plates have rough, bumpy surfaces. The brass heads often develop cracks.

passing through it to provide fast cooling to the system. Figure 2 shows broken pieces of the graphite electrode, the surface damaged brass plate and a cracked brass head unit. Figure 3 shows the three main components as they are arranged in the press and utilized during the actual production process.

In the press, dissimilar materials are in contact with one another at extremely high temperatures

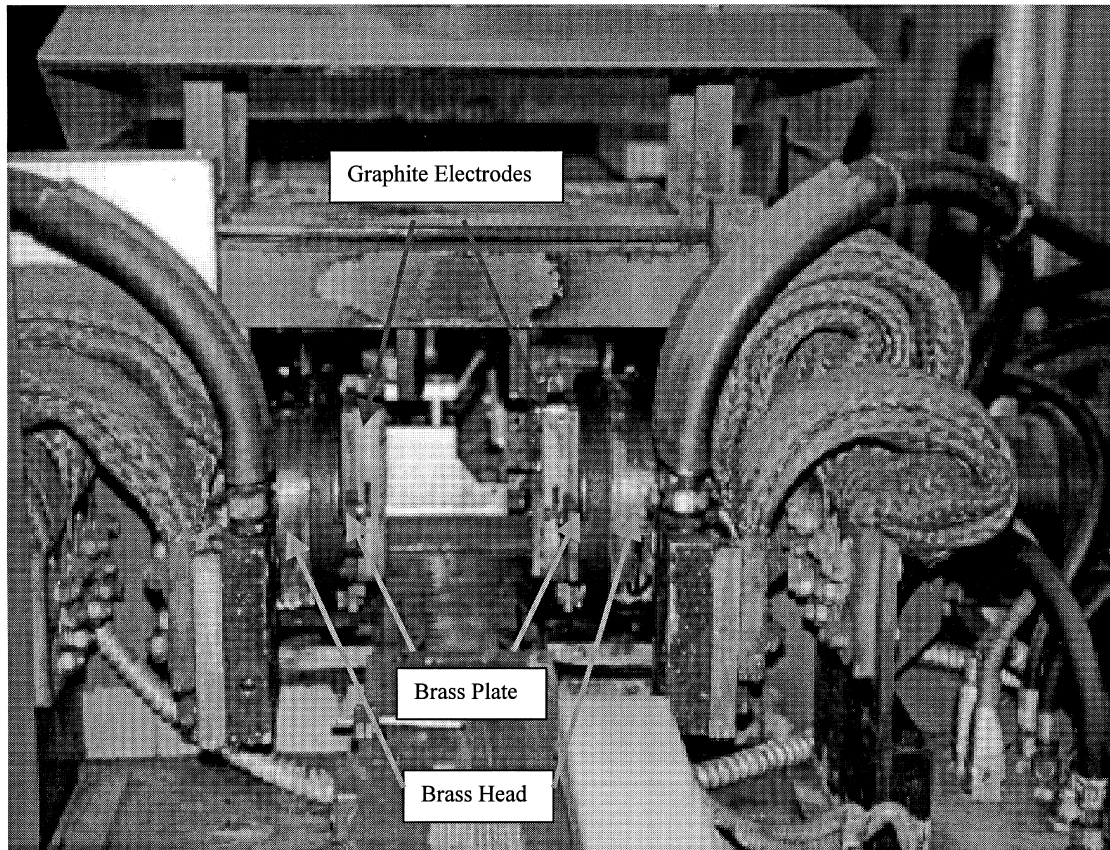


Fig. 3. Actual layout of the graphite electrode, brass plate, and brass head in the press.

and pressures. With repetitive use, as is normal in a production line environment, the graphite is disintegrating and the brass head unit and plate are developing cracks. The graphite is Schunk FE 679 and the electrodes, machined to specification, cost approximately \$25 each. The brass plates are 7-in. diameter discs about $\frac{1}{4}$ in. thick, which can be bought at the prevailing brass rate and machined without much additional cost. The copper 36,000 heads are 7 in. in diameter and 2 in. thick, and after processing to the exact specifications of the tooling, cost \$560 each.

3. Analysis

The three components in front of the hydraulic cylinder, Fig. 4, were monitored while on line. Initially, it was suspected that pressure caused by unequal thermal expansion of the components was being monitored incorrectly. This idea was quickly abandoned when it was found that the pressure measuring devices were located on the proper, hydraulic fluid side of the press [1].

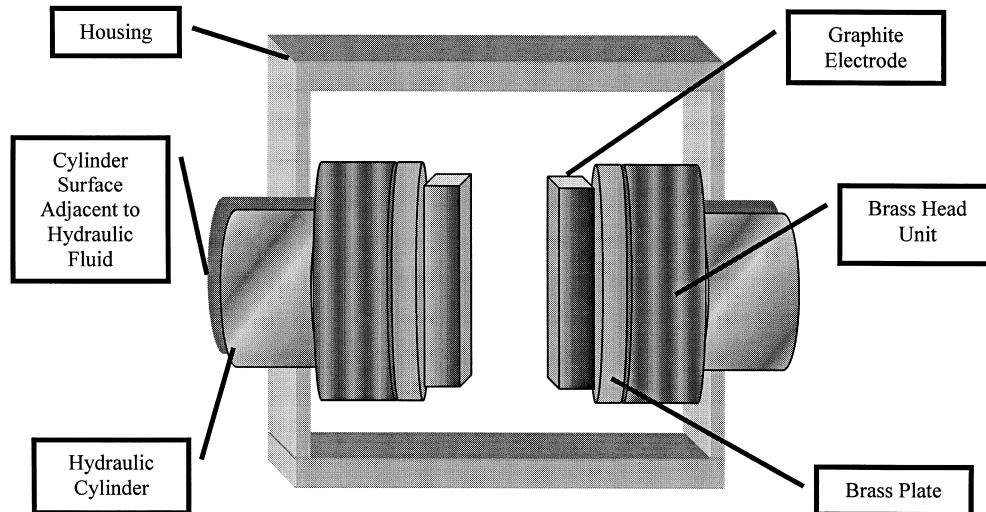


Fig. 4. Housing, hydraulic cylinders, brass heads, brass plates, and graphite electrodes.

3.1. Graphite electrodes

When the electrode samples were removed from the machine, they were clearly marked as coming from station 1 or 2, and from the back or the front. As more samples were collected, it was seen that more electrodes were breaking apart in station 2 than in station 1. For every 10 pairs of electrodes discarded from station 2 there was 1 pair discarded from station 1. This was reasonable because station 1 was a dewaxing station whose temperature does not exceed temperatures of 500°C.

During the study, out of the 25 samples collected only two were broken into four pieces. The rest of the electrodes were intact but were ejected from the machine because of wear around the edges. After further investigation, it was found that within a week the brass plates had to be replaced, and that if the warpage of the brass plates could be eliminated, fracture of the brass head and graphite electrodes would be eliminated.

The focus for the electrode analysis became reducing edge wear instead of eliminating fractures. Microscopic examination indicated that the edge wear was due to oxidation from water vapor, which escapes the brass head during the high temperature operations. For the graphite electrodes, it was decided to run three test pairs of electrodes for two days at a time. One pair was the control set, run under standard operating parameters (7 l/h nitrogen flow rate). To assess how much of an effect the inert gas flow had on electrode oxidation, another pair was run with twice (14 l/h) the normal operating nitrogen gas flow rate. The last set was run with electrode edges coated with boron nitride. A thin ($\frac{1}{8}$ – $\frac{1}{16}$ in.) layer of boron nitride coating covered the outer edges of the pair of electrodes as dimensioned in Fig. 5. Figure 5 shows the pieces F and B (Front and Back) with the mold package contact surfaces facing the viewer. The top and one side of the electrodes have a narrower coating than the bottom and opposite side, in accordance with actual oxidation layers found on almost all discarded electrodes.

Table 1 shows the weights of the control, boron nitride coated and excess inert gas flow rate

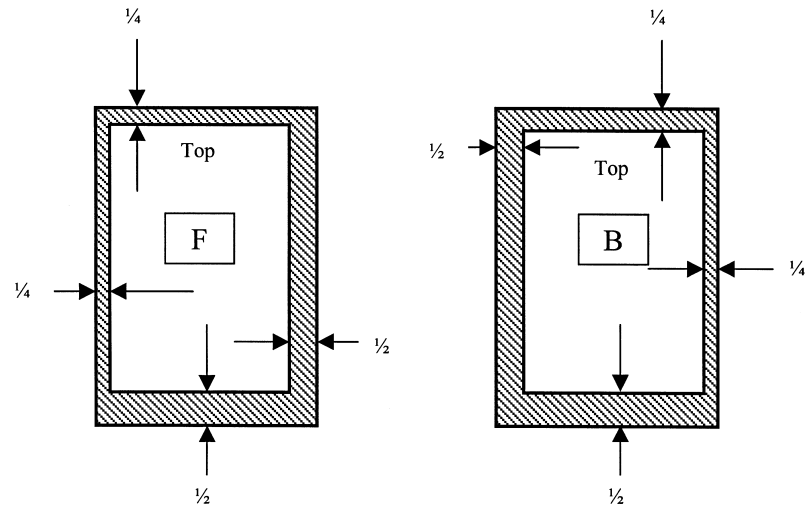


Fig. 5. The dimensions for coating of the electrodes are shown in inches (in.). The electrode surface in contact with blade segments is shown.

Table 1
Graphite electrode weights before and after two days of service

	Electrode #	Weight (g)				Percent weight loss
		Average				
<i>Before service</i>						
After coating	1	1205.11	1205.11	1205.11	1205.11	
	2	1200.79	1200.78	1200.79	1200.79	
	3	1207.66	1207.66	1207.65	1207.66	
	4	1203.62	1203.62	1203.63	1203.62	
	3	1207.95	1207.95	1207.94	1207.95	
	4	1203.89	1203.88	1203.89	1203.89	
	5	1203.44	1203.44	1203.46	1203.45	
	6	1203.62	1203.62	1203.63	1203.62	
<i>After service</i>						
Control set	1	1200.71	1200.70	1200.70	1200.70	0.366
Boron nitride	2	1194.36	1194.36	1194.37	1194.36	0.535
	3	1203.84	1203.85	1203.85	1203.85	0.339
Twice N ₂ flow rate	4	1195.43	1195.43	1195.43	1195.43	0.703
	5	1196.45	1196.45	1196.46	1196.45	0.582
	6	1191.64	1191.62	1191.64	1191.63	0.996

recipient electrodes before and after the two days of production operation. Each sample was weighed three times and an average reading was recorded. Two decimal place accuracy was retained to account for minute changes in weight. It was noted that all weight reduction or oxidation was less than 1%. The data indicate more wear and oxidation on the boron nitride coated electrodes and double inert gas flow rate electrodes than the control pair which received normal service practices. However, what the data fail to show was that the control pair of electrodes were installed later in the day and saw less production time (less than two days) in the press than the boron nitride coated pair or the pair used under twice the normal nitrogen flow. Not having a truly controlled test atmosphere, variables such as the number of mold packages pressed and the type of blade segments joined had a significant impact on the analysis.

3.2. Brass plates

For further information about the brass plates and head units the manufacturing supervisor was consulted [2]. The brand new brass plates start out with a 350/1000 in. thickness. After wear due to production, some damage is incurred. The surfaces of the plates become bumpy and indented, and the plates become warped. The front and back surfaces of the plate are no longer parallel to each other. The plates generally last one week before needing replacement, which is approximately the same time after which the electrodes begin cracking. After a week, the back and front faces of the brass plates are resurfaced, removing a total of 25/1000 in. of material. The resurfaced plates last a day less than in the previous cycle. The plates go through this cycle of resurfacing and replacement into service until they reach a critical thickness of 200/1000 in. At this point they can no longer be used in the press [2].

To test the amount of warping over a period of service time, the surface deviations were measured by a height gage over a service period of two, three, and five days. Table 2 shows the deviation from the center point of each plate at 10 locations, labeled with appropriate dimensions in Fig. 6. It was seen that tremendous warping occurs just after two days of operation. The locations near the outer edge (1, 4, 5, 8, 9, and 10) had highest values of deviations for the two- and three-day runs. For the test samples used for five days (Plates 5 and 6), all throughout the plate there were significant deviations. The signs denote the direction of deviation from the surface in contact with the electrodes. Positive deviation indicates warping toward the brass head; negative deviation, toward the electrode.

3.3. Brass heads

The brass heads are hollowed out to allow a flow of water to cool off the mold package and graphite electrodes. The brass heads on station 1 (dewaxing station) last approximately 5–6 months, while the brass heads on station 2 last 3–4 months. Generally, the surface in contact with the plate is resurfaced three times over a lifetime. The first resurfacing removes 15/1000 in., the second, 25/1000 in., and the third, 30/1000 in. The surface in contact with the hydraulic cylinder (Fig. 4) is resurfaced once over the life of the head. A head will typically last five weeks before initial resurfacing, four weeks after one resurfacing, and about three weeks the last time [2].

Table 2
Surface deviations in plate thickness at 10 locations across plate

Location		1	2	3	4	5	6	7	8	9	10
Run time	Plate number	<i>Before time in service (0.01 in.)</i>									
2 days	1	−2	−2	1.5	2.5	0	8	−1	−1	1	0
2 days	2	2	2	1	2.5	1	0.25	3	6	6	1
3 days	3	2	0	1	2	3	0.5	1	1.5	2	3
3 days	4	3.5	0.5	0	−1	4	3	−0.5	0	−1	5.5
5 days	5	2	0	1	2	−0.5	0	1	3.5	3	2.5
5 days	6	3	1	0	1	4	1	0	0	0	3
Run time	Plate number	<i>After time in service (0.01 in.)</i>									
2 days	1	−13	−7	−1	−4	−5	−1.5	−4	−10	−7	−10
2 days	2	−12	−4	−1	−7	−7	−7	6	−5	−5	−10
3 days	3	14	1	5	14	15	1	4	12	11.5	14
3 days	4	9	1	4	10	13	4	3	11	7	10
5 days	5	5	1	2	7	1	−2	0	6	4	3
5 days	6	14	3	0	8	8	1.5	2	7	8	12

All measurements are in 0.01 in. The center of the discs was used as the zero deviation point.

4. Summary

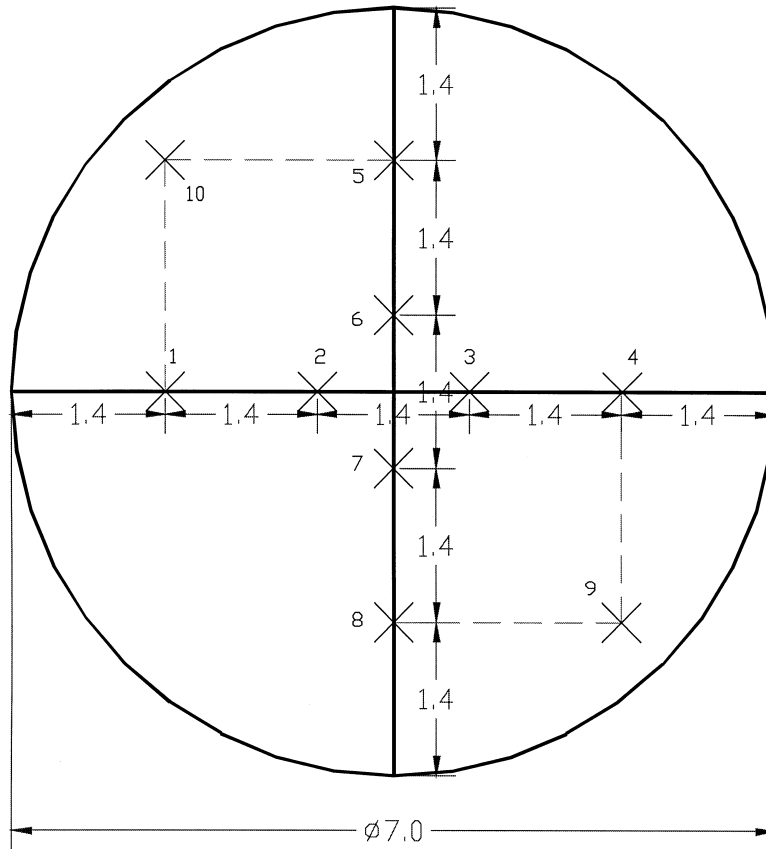
It appears that the failures are occurring in a domino effect. The electrodes oxidize and warp from water vapor in their environment. When pressed in production, this causes the brass plates to warp and incur surface damage. Once the plates are damaged they convey the warpage onto the brass head units.

All of the electrodes tested showed significant wear. A more in-depth analysis of electrode wear in a completely controlled production environment could not be performed. For example, during this study the amount of production work was not equal among electrode pairs and could not be controlled. Actual service life was dictated by the orders processed on the day of the test. The plates show significant warping even after two days.

5. Recommendations

It is recommended that the manufacturer implement the following procedures to reduce component failures:

- (1) The electrodes should be coated with boron nitride, as shown in Fig. 5.
- (2) The brass plates should be resurfaced with no more than two days of service time when the



production has been intensive, and with no more than five days of service time if production has been light.

- ## References

- [1] Sondermaschinen Fritsch. Pressure Sintering Press SPM 75-S Operating Instructions.
- [2] Private communication with Grajeda, Luis J., Diamond Tool Manufacturing Supervisor, 1998.